



The Influence of Plants on the Migration and Transformation of Nitrogen in Plant-Soil Systems: a Review

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Abstract

Nitrogen (N) fertilizer application and atmospheric wet deposition have led to excessive N input into the soil and nitrate leaching into groundwater, which has become a widespread environmental concern. Plants serve as key participants in the soil N cycle. In this review, we critically examine recent advances and knowledge gaps with respect to the role of plants on N migration and transformation in plant-soil systems. The key aspects regarding the influence mechanism and interdependent relationships among plant roots, plant metabolism, plant root exudates, and rhizosphere microbes on N migration and transformation are summarized. Our review reveals that there is a plethora of information regarding the influence of microorganisms on the migration and transformation of N; there is limited knowledge on the effects of different plant root characteristics and root exudates on nitrogen transport and transformation. There are major knowledge gaps regarding quantitative pathways of N migration and transformation by deep-rooted plants to abate negative environmental impacts caused by excess N leaching as well. Environmental stresses, particularly the comprehensive influence of multiple environmental factors on plant-soil systems, need to be further researched to more effectively manage soil and plants.

Keywords Metabolism · Exudates · Residue · Rhizosphere · Microorganism · Environmental stresses

1 Introduction

Nitrogen (N) is an essential component of all living organisms and the main nutrient limiting life on our planet (Kuypers et al. 2018). Some key human activities that affect the cycling of N include fertilizer application, emissions of ammonia (NH₃) and combustion-related nitrogen oxides (NO_x), and discharge of industrial and municipal wastewater (Fowler et al. 2013; Ponette-González et al. 2017; Xin et al. 2019). N fertilizers have been widely used in agriculture to increase crop yield and meet food demands for population

growth, but excessive use has negatively impacted nonpoint source pollution (Guo et al. 2020a, b). Studies have suggested that only ~40% of fertilizer N is taken up by plants (Alva et al. 2006) while more than 30% of fertilizer N was lost through nitrate (NO₃⁻) leaching (Wang et al. 2019). N from fossil fuel combustion and sewage discharge into the water causes NO₃⁻ and ammonia (NH₄⁺) pollution of water bodies (Fowler et al. 2013; Shortle et al. 2020). Yu et al. (2019) reported that the rates (14.5 ± 3.1 megatons of N per year) of anthropogenic N (particularly NO₃⁻ and NH₄⁺) discharged into water exceed the estimated “safe” N discharge threshold by 2.7 times (5.2 ± 0.7 megatons of N per year). In order to solve this problem, Van Dreht et al. (2009) used economic growth and wastewater treatment installations to estimate global N emissions resulting from sewage discharge during the period 1970–2050. They predicted a rapid increase of N per year from 6.4 Tg in 2000 to 12.0–15.5 Tg in 2050 for global sewage emissions. van Puijenbroek et al. (2019) supported this prediction through showing that the N discharge to surface water increased from 5.2 to 10.4 Tg year⁻¹, respectively, between 1970 and 2010.

The total nitrogen in soil includes both organic and inorganic N. Inorganic N consists of mineral N (mainly

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NH_4^+ -N and NO_3^- -N) and fixed ammonium (Kazakis et al. 2020; Ryu et al. 2021). Fixed ammonium is held in soil in its nonexchangeable form; its amount is related to the kind and amount of clay minerals present (Stevenson and Dhariwal 1959). NH_4^+ stems from the mineralization of organic N, N fertilizer application, and release of fixed ammonium (De Boer et al. 2016). It is fixed by negatively charged soil colloids and available to be uptaken and utilized by many plants as a preferred N source (Vranova et al. 2012). Sources for NO_3^- are N fertilizer application and NH_4^+ ; the latter is crucial for the nitrification of microbes under aerobic conditions (Castellano-Hinojosa et al. 2020). NO_3^- has a high migration rate due to its lower absorption properties in soil.

As a major contributor to the N cycle in the plant-soil system, plants have a significant effect on the migration and transformation of N (He et al. 2020b). N cycle is mainly influenced by plants in three ways: preferential flow generated by plant roots, plant metabolism, and plant rhizosphere (Gold et al. 2019). With regard to preferential flow generated by plant roots, Kristensen and Thorup-Kristensen (2004) concluded that compared to non-arable systems, leaching of NO_3^- from cropping systems occurs at a higher rate. Generally, plant assimilation and rhizosphere microorganisms work together as a cohesive unit. For example, Fan et al. (2019) indicated nearly 50% of residual NH_4^+ -N in soil was removed by plant assimilation and nitrification. It is estimated that N input through biological fixation is approximately 52–130 Tg N year⁻¹, of which about 20–60 Tg N year⁻¹ are fixed by crops (Davies-Barnard and Friedlingstein 2020; Soumare et al. 2020). Returning plant residues to the soil provides an alternative and sustainable way to apply N fertilization. For example, Shindo and Nishio (2005) found that the remineralization rates of N after the addition of wheat straw into the soil were 0.71, 0.55, and 0.29 mg N kg⁻¹ day⁻¹ after 7, 28, and 54 days, respectively. Therefore, soil microbial diversity (Guo et al. 2020b; Liu et al. 2020b), plant types (Gao et al. 2020), plant residue (Qin et al. 2019), and other biological factors play crucial roles in the N cycle (Legay et al. 2020; Locke et al. 2018; Rovira 1965; Sun et al. 2016b).

In this review, we address three subjects: (i) effects of plant root characteristics on N migration through the generation of preferential inflow in soil; (ii) mechanisms and influencing factors of plant metabolism, rhizosphere exudates, and rhizosphere microbes on soil N transformation; and (iii) potential drawbacks and challenges of N migration and transformation processes in plant-soil systems. The combination of these three subjects will help shed light on the underlying mechanism and influencing factors of N migration and transformation in plant-soil systems, allowing for the ablation and prevention of N-related environmental and ecological problems.

2 Role of Plants in Nitrogen Migration in the Plant-Soil System

2.1 The Influence of Plant Root on Preferential Flow

The preferential flow path, often associated with well-connected macropore (> 300 μm Ø), significantly affects the vertical migration of water and solutes in the soil (Franklin et al. 2021; Martinez et al. 2021). The preferential flow refers to the flow of water and solutes that bypass the soil matrix and migrate rapidly through soil macropores (Fishkis et al. 2020; Zhang et al. 2021).

Plant roots mainly affect the pore structure in soil and generate preferential flow (Galdos et al. 2020). There are three main ways for plant roots to form soil macropores and accelerate the preferential flow: (i) create cracks in soil through mechanical actions such as interleaving and segmentation, thereby increasing the number of large pores in soil (Liu et al. 2019; Wu et al. 2016, 2017a); (ii) change soil structure by increasing the content of soil organic matter and water-stable aggregates (Leung et al. 2015; Wu et al. 2020); and (iii) seasonal death of the fine roots (root diameter < 1 mm) and the contraction of the coarse roots that cause large pore structures to form in the soil (Mitchell et al. 1995). Studies have shown that some root characteristics, such as root length and root diameter, affect preferential flow in soils (Franklin et al. 2021; Guo et al. 2019). For example, decayed and coarse roots (> 5 mm) generate preferential flow easily (Huan et al. 2020; Nakagawa et al. 2019). This is because these decayed and coarse roots readily form macropores in soils by interpenetrating and shrinking, strongly influencing soil properties, such as hydraulics. For example, Guo et al. (2019) concluded that the preferential flow of decayed roots of *M. sativa* increases the stable infiltration rate of soil by 31.80%. Guo et al. (2020c) indicated that the coarse roots (> 5 mm) form belowground funneling of stem flow can transport water 2.8 m on a downhill slope from the research tree in 30 min.

In addition to the root characteristics, the amount of preferential flow produced by the root system is mainly affected by soil properties and environmental conditions. The main influence factor is soil texture. Sandy loam is more likely to form preferential paths for plant roots in comparison to clay loam (Jingxuan et al. 2018). This is because the greater the mass fraction of sand particles in the soil, the more easily the connected pore structures in the soil form, allowing for uninhibited passage and migration of N. The decrease of soil moisture could promote the generation of preferential flow and its accompanying solute migration. By applying Brilliant Blue dye tracer to soil profiles at high and low antecedent soil moisture, Hardie et al. (2011) investigated

the effect of soil moisture status on preferential flow and concluded that the permeability of the dye tracer in the dry treatment was 9.6 times that in the wet treatment. Seasonal factors are another area that contributes to changing soil structure through their influence on plant growth (Franklin et al. 2021). For example, it is easier to form preferential flow in autumn and winter than in spring and summer. Stumpp and Maloszewski (2010) found that the rate of preferential flow within the soil decreased in August and marginally increased in September in correspondence to changes in plant growth. Fishkis et al. (2020) used the Hydrus-1D transient model to show that the rate of preferential flow increases 13–30% during winter. Lastly, it is worth noting that soil moisture has a significant influence on water flow.

2.2 The Influence of Preferential Flow on N Migration

As a soil physical process, preferential flow is being continuously studied (Lu et al. 2020; Galdos et al. 2020; Nakagawa et al. 2019), but the influence of preferential flow on the N migration is still largely unknown. There are three main pathways for soil N migration: (i) migration into groundwater through underground leaching; (ii) migration into surface water bodies through surface runoff or soil erosion; and (iii) release into the atmosphere through diffusion or gaseous exchange (Gold et al. 2019). Plants form macropores in soils through their root systems, resulting in the generation of the preferential flow paths, which often contain significantly greater concentrations of N as compared to non-flow path regions and accelerate the migration of N, leading to N pollution in groundwater (Franklin et al. 2021; Galdos et al. 2020; Zhang et al. 2020; Nakhforoosh et al. 2021). An example of this phenomenon was studied by Prado et al. (2011). They concluded that preferential flow, coupled with high rainfall rates, can intensify NO_3^- migration at a depth of 1 m under the root zone. Research conducted by Wang et al. (2013) also found that the priority flow (43.8% of the total flow) causes 97.60% of cumulative nitrogen leaching. For agricultural lands, the preferential flow in the soil increases the possibility of nitrogen directly polluting the groundwater and reduces the retention of N around the plant root system, as well as inhibiting fertilizer utilization as a result (Fishkis et al. 2020). Galdos et al. (2020) explored how soil structure was modified by the roots of maize (*Zea mays*), palisade grass (*Brachiaria brizantha* cv. Marandu), and ruzigrass (*Brachiaria ruziziensis*) and how the roots of these plants affected NO_3^- leaching and retention. Galdos et al. (2020) concluded that plants can increase rates of NO_3^- transport because root systems are beneficial for preferential flow. We summarize the effects of plant roots on soil structure and preferential flow on nitrogen migration in Fig. 1.

Various methods are being studied to solve the preferential flow problem in the soil in relation to N pollution and leaching, including the use of enhanced-efficiency fertilizers and refinements in farming practices. For example, Miller et al. (2020) developed an HYDRUS-1D solute transport model that precisely assesses the effectiveness of various kinds of N application methods to help abate N pollution in soil. Wang et al. (2002) proved that compared to conventional fertilization, whole source fertilization can effectively solve the nitrogen leakage loss along preferential flow paths. In addition, the land use change from intensive arable to semi-intensive grassland systems or forestation is a powerful measure for reducing nitrate leaching that occurs as a result of preferential flow mechanics on a landscape scale (Biernat et al. 2020; Zhu et al. 2019).

Notably, after the establishment of preferential flow paths, localized high rates of microbial activity can develop rapidly due to the local accumulation of nutrients, affecting N transformation (Franklin et al. 2021). Plants significantly influence the N transformation process as well through their own metabolism, root exudates, and other means.

3 Role of Plants in Nitrogen Transformation in the Plant-Soil System

In addition to the plant roots in nitrogen migration in the plant-soil system, two main processes were found to influence the soil nitrogen transformation pathways in the plant-soil system by plants, including plant metabolism (plant assimilation and return of plant residues) and the plant root exudates. Root exudates in the rhizosphere mediate microbial processes that can drive soil nitrogen transformation and increase nitrogen availability (Liu et al. 2022), while microbes and N availability generate bidirectional responses that influence root exudate chemistry (Carvalhais et al. 2011; Smercina et al. 2021). Therefore, it is necessary to understand the interdependent relationships among plant metabolism, rhizosphere-microorganisms' interactions, and plant root exudates to improve plant N acquisition and minimize N inputs and losses in sustainable agriculture.

3.1 The Influence of Plant Metabolism on the Process of Nitrogen Transformation

3.1.1 The Role of Plant Assimilation in Nitrogen Transformation

Plant assimilation serves as an important contributor to the elimination process of residual NH_4^+ and NO_3^- in soil (Liu et al. 2020a; Tian et al. 2020). In order to understand how plants influence mineral N transformations to reduce NO_3^- leaching, the mechanism of plant assimilation of the

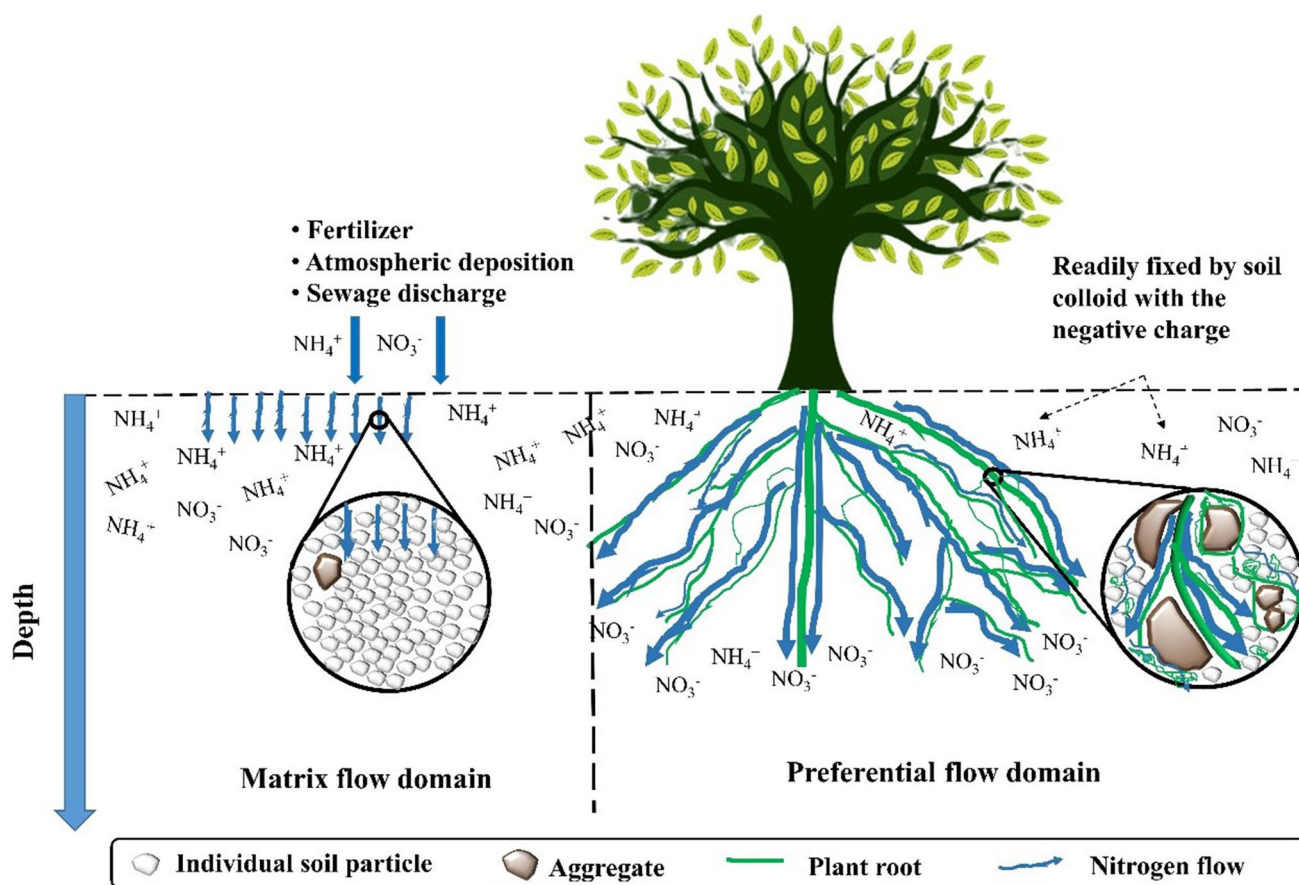


Fig. 1 The effect of plant roots on preferential flow in terms of N migration

mineral N and the factors that influence this process must be understood. Plants can directly synthesize amino acids after absorbing NH_4^+ ; however, NO_3^- is firstly reduced to NH_4^+ through nitrate reductase and nitrite reductase before being converted into amino acids (Liu et al. 2008; Yoneyama and Suzuki 2019). Although the forms of N uptake by plants are different, they undergo N transformation by enzymes such as glutamine synthase and glutamate synthase to form different amino acids (Carr et al. 2020). Differences between characteristics of NO_3^- and NH_4^+ can significantly impact plants' selective assimilation. For example, although the mobility of NH_4^+ in the soil is not as strong as NO_3^- , the assimilation of NH_4^+ by plants saves energy consumption by 8~17% compared to equivalent NO_3^- assimilation (Tang et al. 2020). Generally, it is economic for plants to absorb NH_4^+ directly. However, for upland crops, NO_3^- will be mainly absorbed (Wang et al. 2020). This is because even if there is a lot of NH_4^+ infiltrated into the soil, NH_4^+ in the soil is nitrified to NO_3^- by microorganisms and then absorbed by upland plants. By tracing of ^{15}N -labeled nitrate and ammonia in roots, Yoneyama and Suzuki (2019) demonstrated that different species of plants absorb and assimilate N through similar methods but in varying

amounts. Figure 2 summarizes the mechanism of plant assimilation for N transformation.

Several factors can affect the assimilation of nitrogen by plants, such as plant biomass, plant species, and the environment. Plants with higher growth rates, broader root systems, and larger biomass can uptake higher amounts of nutrients including N (Meena et al. 2017). Niu et al. (2017) supported this conclusion and suggested that with the increasing age of *Robinia pseudoacacia*, the N content in leaves and branches increases and reaches a peak maximum value at 40 years of age. Plant species with different genes and synthesized enzymes that are involved in primary N assimilation lead to plants having varying absorption ratios of NH_4^+ and NO_3^- (Liu et al. 2018; Tang et al. 2020). For example, some plants such as *Solanum tuberosum* and *Arabidopsis thaliana* are sensitive to NH_4^+ and mainly assimilate NO_3^- whereas some, such as tea and rice plants, are highly adaptable to NH_4^+ and even use it as the sole N source (Andrews et al. 2013). It is worth noting that plants can reach the peak period for nitrogen uptake during the growing season, but exhibit comparatively limited absorption rates during dormant states (Zhao et al. 2010). With regard to environmental

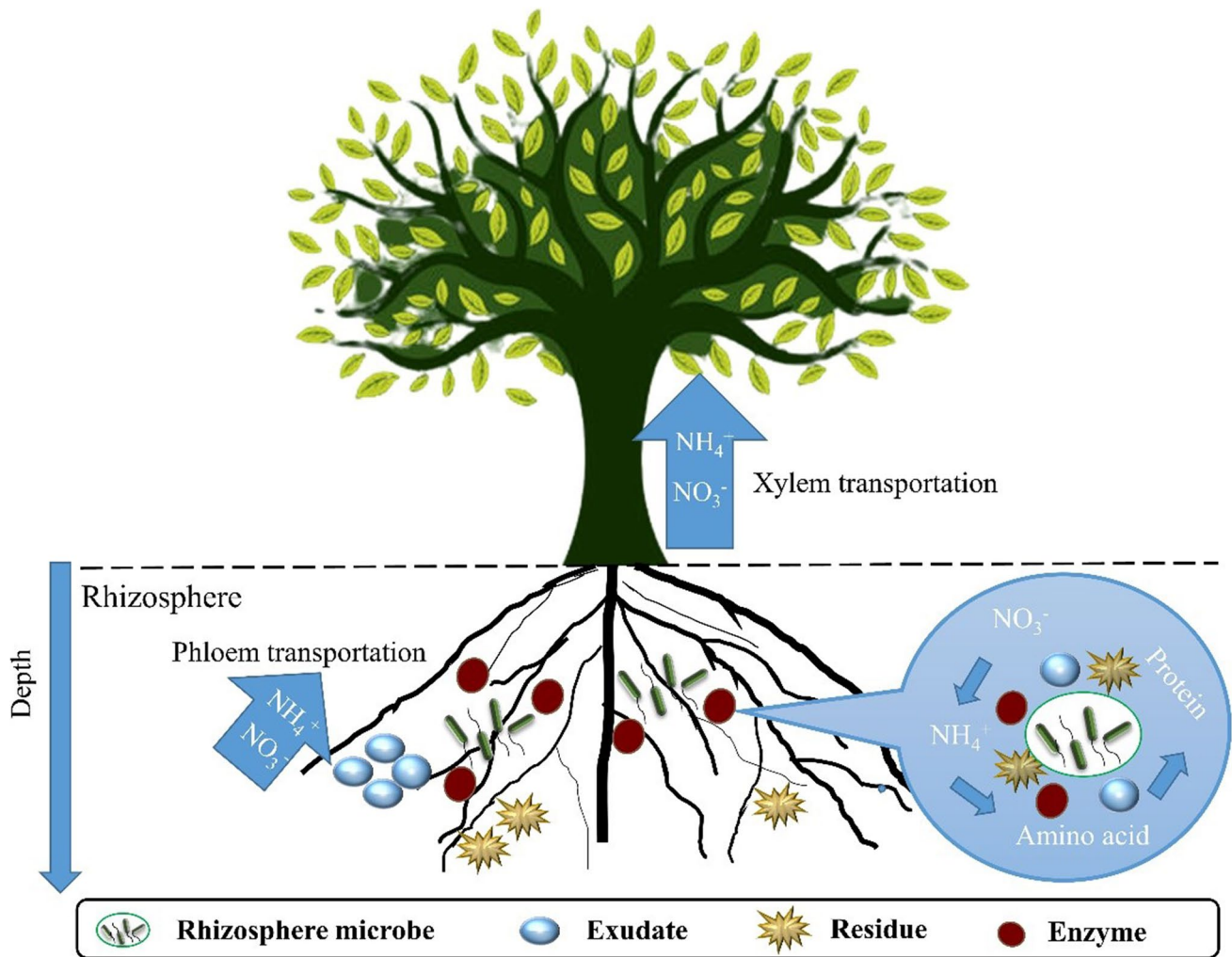


Fig. 2 The effect of plant assimilation on soil in terms of N transformation

factors, N supply is the limiting factor for absorbed N and soil moisture can significantly affect nitrogen assimilation. Heidari and Mohammad (2012) concluded that N content in plants shares a positive correlative relationship with decreasing N levels in soil, specifically from 225 to 75 kg N ha⁻¹. Zhang (2003) found that under the conditions of high light intensity, water deficit, and soil moisture content of 8~15%, a significant increase occurred for root nitrate reductase activity. However, excessive irrigation will reduce the activity of nitrate reductase in plant leaves, severely inhibiting its activities and nitrogen assimilation in plants.

3.1.2 The Role of Plant Residue in Nitrogen Transformation

Plant residues include the leaves, branches, flowers, plant litters, dead roots, residual stubble, and straw after crop harvest (Li et al. 2019; Vityakon et al. 2000). The recovery of plant residues, especially crop residues, is believed to benefit soil fertility and reduce nitrogen loss (Coskun et al.

2017; Haris et al. 2021; Yao et al. 2009). When plant residues are returned to the soil, they can cause soil inorganic N changes through soil organic N mineralization, and plant residue organic N mineralization. Studies have concluded organic nitrogen mineralization from plant residues can increase soil inorganic nitrogen concentrations (Chen et al. 2014). Therefore, agricultural activities increasingly use plant residues in combination with synthetic N fertilizer as the main energy materials for soil microbial activities (Li et al. 2017). Crop residues may be left on the soil surface and affect soil temperature, soil moisture, soil N content, dissolved organic carbon content, and microbial activity, as well as soil N₂O emissions in a complex manner (Li et al. 2019). N₂O, produced naturally in soils through the microbial processes of nitrification and denitrification, contributes to the atmospheric greenhouse effect at a 5% rate, which is second only to CO₂ and CH₄ (Yao et al. 2009). Table 1 displays the effects of returning plant residues to the field on N transformation and N₂O losses in soil.

Table 1 Role of plant residues return to the field on N₂O losses in soil

Plant residue	Composition and quantity of plant residue	Effects of residue return on N ₂ O losses	Reference
Maize	Annual average production was 11.6 Mg ha ⁻¹ with a carbon of 5014 kg C ha ⁻¹ and nitrogen of 96.6 kg N ha ⁻¹	The application of residue reduced N ₂ O emission factors (0.62% and 0.67%, respectively), compared to the fertilizer-only treatment	Li et al. (2019)
Wheat straw	3.75 × 10 ³ kg/ha of wheat straw	On average, straw coverage tended to significantly reduce N ₂ O emissions by about 30%	Ma et al. (2007)
Soybean and corn	Soybean and corn straw with different carbon to nitrogen ratios (75 and 130, respectively)	N ₂ O emissions incorporated with ripe soybean or corn residues are reduced approximately 30% and 80%, respectively, compared to the no-residue control	Pelster et al. (2013)
Wheat straw mulching and maize straw	–	N ₂ O emissions decreased by 46.9% through optimizing N management of the straw return	Huang et al. (2013)
Sweet corn	Carbon to nitrogen ratios of 27 for sweet corn	The surface applied sweet corn residue treatment significantly decreased fertilizer-induced N ₂ O emissions by 22%	Rezaei Rashti et al. (2017)
Sugarcane stalk	The plant residues were oven-dried at 70 °C to a constant weight and ground to pass through a 2-mm stainless steel sieve, with C and N content 472 g kg ⁻¹ and 4 g kg ⁻¹ , respectively	The cumulative N ₂ O emission for the amendment of sugarcane stalk plus urea was approximately 40% lower than that for the sugarcane stalk amendment	Huang et al. (2004)
Crop, legumes and vegetables, cereal residues	Different residues with different carbon to nitrogen ratios	Crop residue amendment generated significantly positive effects on soil N ₂ O emissions, when C:N ratios of crop residues were < 45, slightly positive effects for C:N ratios of 45–100, and slightly negative effects for C:N ratios > 100. Compared with legumes and vegetables, cereal residues generated lower effect size	Chen et al. (2013a)
Crop residue	Crop residue returning in subtropical, tropical, and warm temperate conditions	N ₂ O release was significantly inhibited by 8% under residue returning by plough	Hu et al. (2019)
Organic olive crop weed residues	Residues with the highest polyphenol (2%) and lignin (11%) contents	The 15 N–N ₂ O data indicated that residue quality had no effect on the denitrification of applied fertilizer N	Garcia-Ruiz and Baggs (2007)
Sugarcane	Three residue types: green plant biomass, mature aboveground biomass, straw, and senescent plant biomass	Green plant biomass increased N ₂ O emissions by 89%, and mature aboveground biomass by 40%. Conversely, straw and senescent plant biomass did not increase N ₂ O emissions	Abalos et al. (2022)

Based on the findings from Table 1, the residue return method decreases N_2O emissions by 20–80% and the cumulative emissions of N_2O are negatively correlated with the C:N ratio in plant residues. This may be due to plant residues with higher C:N ratio presumably acting as a stimulant for NH_4^+ immobilization and N_2O consumption through its reduction to N_2 , thereby reducing the risks of significant N_2O emissions (Huang et al. 2013; Rezaei Rashti et al. 2017). For plants, the composition and quantity of plant residues, incorporation timing, and carbon-to-nitrogen ratio of plant residues all have an impact on the dynamic changes of soil N (Pelster et al. 2013; De Almeida et al. 2016; Foltz et al. 2019; Pu et al. 2019). The biochemical composition of plant residues has an impact on N availability, nitrification, and denitrification, influencing levels of N_2O emissions as well. For example, plant residues have higher lignin and polyphenol concentration can be negatively related to N_2O emissions (Garcia-Ruiz and Baggs 2007). On the other hand, some allelochemicals produced by the decomposition of crop residues can further inhibit the production of soil N_2O (Ma et al. 2007; Pelster et al. 2013).

Through these combined findings, it is clear that plant residues have notable ecological significance in reducing the use of fertilizers and curbing the greenhouse effect effectively (Malhi et al. 2006). However, if there is not enough available N in soil with returned plant residues during the plant growth stage (absorb approximately 70–80%) (Chen et al. 2014), the yield will be reduced. Conversely, if there is excessive available N in the soil during the seedling or dormant stage (absorb only a small proportion), N loss may occur (Zhao et al. 2010). Therefore, as research becomes more advanced, some methods such as the use of immobilizers and the locations of the returned plant residues may be identified that allow the return of plant residues to soils to become an efficient method for improving farmland nitrogen dynamics (Luo and Xie 2019).

3.2 The Influence of Plant Rhizosphere on the Process of Nitrogen Transformation

3.2.1 The Influence of Root Exudates on Nitrogen Transformation Microbial Communities

Although plants take up and assimilate both NO_3^- and NH_4^+ , they cannot directly engage in some N transformation processes such as nitrification and denitrification. Plants can control N transformations catalyzed by the fungal and prokaryotic populations in and near the rhizosphere by releasing root exudates (Moreau et al. 2019). Root exudates are the shared language between plants and organisms, acting as energy sources and chemical attractants for microbes as well as a series of primary and secondary plant metabolites, many of which are secreted into the rhizosphere (Du

et al. 2020; van Dam and Bouwmeester 2016). Even the root exudates of a small plant, such as *Arabidopsis thaliana*, may contain more than 100 different metabolites promoting interactions with biotic and abiotic environments (Boccuzzi et al. 2021; Coskun et al. 2017). Although soil microbes significantly drive the global N cycle, plant roots can control the abundance of microorganisms in and near the rhizosphere by releasing root exudates to influence N transformation (Wang et al. 2021). Root exudates oversee four major soil N transformation processes: ammoniation, nitrification, denitrification, and N_2 fixation. Table 2 summarizes the influence of plant root exudates on N transformation in the soil.

Root exudates play various roles (positive or negative) in the N transformation. The negative role of root exudates mainly refers to the use of compounds in root exudates that inhibit nitrification, which is collectively known as biological nitrification inhibitors (Moreau et al. 2019). While root exudates have long been known to control soil nitrification, to date, only five biological nitrification inhibitor compounds have been isolated: sorgoleone, sakuranetin, methyl 3-(4-hydroxyphenyl) brachialactone, and 1,9-decanediol (Table 2). The positive role of root exudates refers to exudates that promote the activity of N cycle-related enzymes or increase the abundance of the relevant microbes. Thus, root exudates can influence the nitrogen transformation process by inducing gene expression of nitrogen-transforming bacteria (Hassan and Mathesius 2012). Growing evidence suggests that rhizosphere exudates serve as an essential strategy that can stimulate the ammoniation process (soil organic N decomposition by microbes) (Bernal et al. 2016; Moreau et al. 2019; Sun et al. 2016b). The reason for this is rhizosphere exudates can release labile C, causing an increased microbial demand for N within rhizosphere microbial community (Huo et al. 2017). As a result, proteases and deaminases release N from N-containing polymers in soil (Moreau et al. 2019). Rhizosphere exudates influence nitrogen fixation in a similar way. Approximately 40% of net fixed organic carbon was allocated belowground (Jones et al. 2009), of which root exudates could transfer about 5–21% of all the photosynthetically fixed nutrients and organic carbon into the rhizosphere (Miao et al. 2020). These nutrients and organic carbon can drive the abundance and diversity of nitrogen-fixing bacteria in soil microhabitats and promote rhizospheric symbiotic N fixation (Li et al. 2016; Haichar et al. 2014). However, it is currently unclear how root exudates influence denitrification. Some biological denitrification inhibitors such as procyanidins can inhibit nitrate reductase and reduce the uptake of NO_3^- by roots (Bardon et al. 2016) while others such as organic acids and soluble sugars can change the scale and structure of denitrifying bacteria community and increase the denitrification rate (Wu et al. 2017b). Therefore, gaining more

Table 2 Role of root exudates in the process of N transformation in the soil

Role of root exudates in the N cycle	The process of N transformation	Root exudates	Plant species	Function in the process of N transformation	Effects of root exudates on N transformation	Reference
Negative	Nitrification	Sorgoleone	<i>Sorghum bicolor</i>	Blocks ammonia monooxygenase (AMO) and hydroxylamine oxidoreductase (HAO); allelopathic compound	The inhibition rate of 9.2 μM Sorgoleone about ammonia monooxygenase (AMO) is 73.7% and hydroxylamine oxidoreductase (HAO) is 76.2%	Subbarao et al. (2013)
Negative	Nitrification	Sakuranetin	<i>Sorghum bicolor</i> ; rice	Blocks AMO and HAO; phytoalexin	Inhibition rate of 1.30 μM Sakuranetin about AMO is 76.0% and HAO is 71.0%	Subbarao et al. (2013)
Negative	Nitrification	Methyl 3-(4-hydroxyphenyl) propionate (MHPP)	<i>Sorghum bicolor</i>	Influence the abundance of ammonia-oxidizing bacteria and archaea	Potential nitrification activity was high (83 $\text{mg NO}_3^- \text{N kg}^{-1} \text{ day}^{-1}$) but decreased by about 80% when treated with MHPP	Nardi et al. (2013)
Negative	Nitrification	Brachialactone	<i>Brachiaria humidicola</i>	Blocks AMO and HAO; reduces field-level nitrification and N_2O emission	Inhibition rate of 5.0 μM brachialactone about AMO is 59.7% and HAO is 37.7%	Subbarao et al. (2009)
Negative	Nitrification	1,9-Decanediol	Rice (<i>Oryza sativa</i>)	Blocks AMO and HAO	Inhibition rate of 100 $\text{ng } \mu\text{l}^{-1}$ of 1,9-decanediol about AMO is 82.49%	Sun et al. (2016b)
Negative	Denitrification	Procyanidins	<i>Fallopia</i> spp.	Allosteric inhibitor of nitrate reductase	At 0.05 mg ml^{-1} procyanidins, denitrification was inhibited by about 65%	Bardon et al. (2016)
Positive	Ammonification	Amino acids and amino sugars	<i>Lobelia pine</i>	Enhanced fast-cycling N degrading enzyme activities	When the C:N ratio of exudates is <7, an increase occurs in the efficiency of SOM decomposition	Meier et al. (2017)
Positive	Ammoniation	Glucose, alanine	<i>Quercus</i>	Alanine produced the greatest excitation effect and increased soil microbial activity	Enzyme activity increased as much as 13 times in the two deeper horizons (1.0–1.3, and 2.7–3.0 m) after the addition of the alanine	Bernal et al. (2016)
Positive	N_2 fixation	Flavonoids	<i>Myrica gale</i>	Rhizobium including <i>Azotobacter brasilensis</i> and <i>Azotobacter stalk</i> , stimulate root colonization through diazo nutrition	Fruit exudates at 1 mg l^{-1} significantly enhanced nitrogen fixation activity of the ACN14a and M16467 Frankia strains	Hassan and Mathesius (2012)
Positive	N_2 fixation	Genistein	maize	Maize root exudates elicited a nearly twofold increase in genistein exudation	Faba bean/maize intercropping enhances productivity, nodule formation, and N_2 fixation of faba bean through interspecific root interactions	Li et al. (2016)
Positive	Denitrification	Organic acids and soluble sugars	<i>Phragmites australis</i> , <i>Typha angustifolia</i>	The abundances of <i>nirK</i> - and <i>nirS</i> -encoding bacteria were significantly influenced by the concentration of glucose and sucrose	The average TOC is 0.1715 to 0.9221 mg g^{-1} root DM day^{-1} , which could maintain a denitrification rate of about 156–841 $\text{kg NO}_3^- \text{N ha}^{-1} \text{ year}^{-1}$	Wu et al. (2017b)
Positive	Denitrification	Oleamide and erucamide	Duckweed	Strengthen denitrification, targeting two main denitrifying enzymes in bacteria, nitrate and nitrite reductases	When 40 mg l^{-1} erucamide was added, the NAR activity under anaerobic and aerobic conditions increased by 44.5% and 25.9%, while the NIR activity increased by 164.6% and 243.0%, respectively	Sun et al. (2016a)

detailed information on how these root exudates shapes the rhizosphere microbiome may assist in better understanding general plant strategies related to plant with micro-organisms to influence N transformation.

Although every plant produces exudates, the composition and quantity of root exudates entering the soil are variable and affected by plant species, age, and climatic conditions (Calvo et al. 2017; Canarini et al. 2019; Pei et al. 2020). As observed in Table 2, exudation is defined by the genotype of the host and changes in accordance with the plant's developmental stage. For example, as *Arabidopsis* ages, the sugar exudation decreases while amino acid and phenolic exudation increase (Chaparro et al. 2014). Furthermore, climate change affects the root exudates and, in turn, nitrogen transformation. Pei et al. (2020) showed a significant increase in three kinds of root exudate levels with a simultaneous decrease in cytokinin concentrations in response to elevated CO₂ levels. Garnier (1995) showed that photosynthetic nitrogen use efficiency is a major determinant of both plant and leaf nitrogen productivities. This can accelerate microbial activities with regard to the decomposition of the N cycle in the rhizosphere, especially under N-limited conditions. Yin et al. (2013) and Shen et al. (2020) found a similar effect at elevated temperatures with variance in plant growth stage and root exudate type. However, it remains unclear how enhanced root exudation and climate change affects these N transformation processes. Therefore, the process of N transformation mediated by root exudates in ecosystems under climate change should still be the focus of future research.

Root exudates may shape the microbiome and N availability, but microbes and N availability can influence root exudate chemistry in return. For example, Zhu et al. (2016) indicated that the N rate was positively correlated with maize root exudation and rhizosphere bacteria abundance. Carvalhais et al. (2011) support this view and concluded that under N deficiency, lower amounts of amino acids were found in maize root exudates. Plants and soil microbes interact closely in the rhizosphere, exchanging functional benefits (Smercina et al. 2021). Root exudates can shift N mineralization and N fixation to influence the supply of plant-available N as well. For example, Bernal et al. (2016) concluded that the alanine of *Quercus* produced the greatest excitation effect for N availability to promote N assimilation by plants and increased soil microbial activity by as much as 13 times. Li et al. (2016) revealed that in faba bean/maize intercropping maize root exudates enhance the productivity, nodulation, and nitrogen fixation of faba bean, which has important implications for developing sustainable agriculture. Therefore, knowledge of such variable contributions of root exudates in relation to N transformation reactions such as N₂ fixation, uptake of mineral N, and N₂O release will help reduce global reliance on fertilizers and their environmental impact.

3.2.2 The Influence of Rhizosphere Microbes on the Process of Nitrogen Transformation

Plants are exposed to trillions of microbes under natural conditions (Chen et al. 2016). These microbes colonize and occupy different areas of soil. Some of these areas provide interfaces with plant roots such as the rhizosphere and rhizoplane while others serve as chambers like the endosphere (considered a secondary genome of the plant) (Kumar and Dubey 2020; Muerdter et al. 2018). Rhizosphere microbes are a group of microbes that are significant for providing various ecosystem services (such as N cycle and carbon absorption) (Adl 2016; Ahkami et al. 2017). Some functional microbes and their genes, which are templates that control enzyme synthesis, involved in the N cycle in the plant-soil system are summarized in Table 3 and Fig. 3.

It can be concluded that plant rhizosphere functional microorganisms and N transformation-related enzymes play important roles in N cycling. Studies have shown that more than 80% of N sources can be utilized by rhizosphere microbes when adding disparate N sources into the rhizosphere (He et al. 2020a; Owen and Jones 2001). Plant rhizosphere functional microbes possess functional genes encoding key enzymes that relate to N-fixation, ammoniation, nitrification, and denitrification, etc., which can significantly stimulate the gross rates of these N transformation processes. As a result, these functional genes serve as reliable predictors of soil N transformation (Hu et al. 2020b). A variety of microorganisms with the *nifH* operon, such as *arbuscular mycorrhizal fungi*, *ectomycorrhizal fungi*, and N-fixing bacteria, have the ability to convert atmospheric N₂ into ammonia by establishing mutualism interactions with plants. Soybean legumes, which serve as one of the highest contributors to biological nitrogen fixation, engage in highly specific mutualistic associations (culminates in the formation of nodules) with soil bacteria and annually fix about 16.4 million tons of N per year (Hungria and Mendes 2015). Lowman et al. (2016) indicated that several *Burkholderia* species can fix atmospheric nitrogen due to possessing the *nifH* operon, allowing it to act similarly to *B. vietnamiensis*, a known free-living nitrogen-fixing bacterium.

Ammonization process is a critical step in the supply of nitrogen available to plants. It is presently known that the predominant proteases of bacterial origin are mainly alkaline metalloproteases (*apr*), neutral metalloproteases (*npr*), and serine proteases (*sub*) (Bach et al. 2001), which decompose various nitrogen-containing organics into polypeptides, amino acids, and ammonia (NH₃) extracellularly to participate in the ammoniation process. Asmar et al. (1994) found that the activity of the extracellular enzymes was highly correlated with the rates of net N solubilization, which seems to suggest that extracellular enzyme activity characterized by proteases is a limiting factor for soil organic

Table 3 Some rhizosphere functional microbes and genes from different host plants that involved in N transformation

N transformation processes	Rhizosphere functional microbes (only some microbes related to N cycling reported in the reference are list)	Microbial functional genes	Plant species	Content or effect of rhizosphere functional microbes	References
N-fixation	Diazotrophic (<i>Rhizobium</i> , <i>Sinorhizobium</i> and <i>Azospirillum</i>)	<i>nifH</i>	<i>Douglas-fir</i>	The abundance of <i>nifH</i> positively correlated with total C, organic C, and N content	Levy-Booth et al. (2014)
	<i>Azospirillum brasilense</i> , <i>Klebsiella</i>	<i>nifH</i>	<i>Pennisetum</i>	The number of diazotrophic bacterial ranged from 10^2 to 10^6 bacteria g^{-1} fresh tissues of both genotypes	Videira et al. (2012)
	<i>Klebsiella</i> , <i>Pseudomonas</i>	<i>nifH</i>	<i>Sugarcane</i>	The max identity of <i>Klebsiella</i> from Brazilian <i>sugarcane</i> was 100–263% identical	Magnani et al. (2010)
	<i>Pseudomonas</i> , <i>Azospirillum</i> , <i>Rhizobium</i>	<i>nifH</i>	<i>Lasiurus sindicus</i>	<i>Pseudomonas</i> -related <i>nifH</i> genes may be important along with other genes such as <i>Azospirillum Brasilense</i> , and <i>Rhizobium</i> sp.	Chowdhury et al. (2009)
	<i>Rhizobium</i> , rare fungi <i>Kurtzmanomyces</i>	<i>nifH</i>	Wheat (<i>Triticum aestivum</i> L.)	The abundance of the bacterial genus <i>Rhizobium</i> was the highest in rhizosphere soil but was absent in bulk soils	Essel et al. (2019)
Ammonification	Arbuscular mycorrhizal fungi and ectomycorrhizal fungi	<i>nifH</i>	<i>Leymus secalinus</i>	The relative abundance of <i>Basidiomycota</i> increased with N levels	Liao et al. (2021)
	Diazotrophic (<i>Rhizobium</i> , <i>Sinorhizobium</i> and <i>Azospirillum</i>)	<i>nifH</i>	<i>Douglas-fir</i>	The abundance of <i>nifH</i> positively correlated with total C, organic C, and N content	Levy-Booth et al. (2014)
	<i>Bacillus</i>	<i>npr,sub</i>	<i>Para grass (Urochloa mutica)</i>	25% isolates of histoplane fraction were members of <i>Firmicutes</i> and <i>Bacillus</i> (8.33%) and others represented the group of <i>Firmicutes</i>	Mukhtar et al. (2016)
Nitrification	<i>Bacillus</i>	<i>npr,sub</i>	<i>Soybean (Glycine max)</i>	Amounts of <i>Bradyrhizobium</i> (rank 4), and <i>Bacillus</i> (rank 21 and 170) significantly increased during the flowering stage	Sugiyama et al. (2014)
	<i>Pseudomonas</i>	<i>apr</i>	Wheat <i>Triticum aestivum</i>	Among bacterial isolates from the wheat rhizosphere	Majeed et al. (2015)
	<i>Nitrosomonas</i> , <i>Nitrosospira</i>	<i>amoA</i>	<i>Typha orientalis</i>	1.45% <i>nitrospirae</i> bacteria groups were in the rhizosphere of <i>Typha orientalis</i>	Shu (2015)
	<i>Nitrococcus</i>	<i>nor</i>	<i>Halocnemum strobilaceum</i> ; <i>Phragmites australis</i> ; <i>Kareliniacaspia</i>	The abundance of <i>Nitrococcus</i> was 2.91×10^5 – 1.05×10^6 copies/g soil, which was the highest in the rhizosphere of <i>Phragmites australis</i>	He (2017)

Table 3 (continued)

N transformation processes	Rhizosphere functional microbes (only some microbes related to N cycling reported in the reference are list)	Microbial functional genes	Plant species	Content or effect of rhizosphere functional microbes	References
Denitrification	<i>Herbaspirillum</i> , <i>Pseudomonas</i>	<i>napA</i> , <i>narG</i> , <i>nirK</i> ,	<i>Potamogeton crispus</i>	The mole ratio of $\text{NH}_4^+/\text{NO}_3^-$ had a strong, positive correlation with the abundance of denitrifying bacteria	Hu et al. (2020)
	<i>Herbaspirillum</i> , <i>Pseudomonas</i>	<i>nirS</i> , <i>norB</i> , <i>nosZ</i>	Paddy	More than 40% of <i>nirS</i> OTUs were affiliated with <i>Herbaspirillum</i> , a key N-cycle player in paddy soil	Duan et al. (2018)
	<i>Neisseriales</i> , <i>Rhodocyclales</i> , and <i>Burkholderiales</i>		Rice paddy	The amount of 16S rRNA and <i>nirS</i> transcripts increased along with denitrification activity	Yoshida et al. (2012)
Anammox	<i>Jettenia</i> , <i>Brocadia</i> , and <i>uenenia</i>	<i>hzo</i>	<i>Hydrilla verticillata</i> ; <i>Potamogeton maackianus</i>	The highest diversity of the <i>anammox</i> community was found in the <i>Potamogeton maackianus</i> rhizosphere	Yin et al. (2020)

nitrogen dissolution. It is noted that protease excretion also is a widespread physiological trait in some ectomycorrhizal fungi, such as *amanita muscaria*, *russula chloroides*, and *lactarius* (Nygren et al. 2007).

Nitrification and denitrification are the microbial processes primarily responsible for transformation of mineral N in soil (Moreau et al. 2019). By detecting the abundance of nitrification and denitrification genes or enzyme activities, it can effectively reflect the reaction intensity of nitrification and denitrification in soil. For example, Li et al. (2020) concluded that the relative abundance of *amoA* of ammonia-oxidizing bacteria serves as a more optimal predictor for Net N nitrification rates; the relative abundance of *ureC* and *nirK* of nitrite reductase and urease for net mineralization rates; and the relative abundance of *narG* and *nirS* of denitrifying bacteria for N_2O emissions rates. Chen et al. (2013b) used quantitative PCR to study the changes of nitrification (*amoA*) and denitrification genes (*nirS*, *nirK*, *norB*, and *nosZ*) in a bioretention system, indicating that the abundance of denitrification genes is directly proportional to nitrate removal. Ammonium oxidation is a late-discovered microbial-mediated transformation process of nitrogen-containing compounds in the ecosystem. The *hzo* gene as a marker gene is of great significance to study the community characteristics of anammox bacteria in natural ecosystems (Kuypers et al. 2018).

Recent studies indicated that the environmental conditions of microbes involve the supply of carbon (C) (Perthame et al. 2020), N (Srivastava et al. 2020; Yuan et al. 2020), and oxygen (Chunmei et al. 2020), which are the main factors that control the rate of N transformation. C and N are energy substances for microbial growth; the ratio of them has been extensively studied as the main factor affecting the N transformation processes such as N mineralization, denitrification, and its products (Li et al. 2014; Wang et al. 2017; Zumft 1997). Organic materials with low carbon-to-nitrogen ratio ($\text{C/N} < 25$) have a higher nitrogen mineralization rate than those with a high carbon-to-nitrogen ratio ($\text{C/N} > 25$). This is because C limits microbe growth under low C/N conditions, resulting in minimal N fixation due to the presence of sufficient N sources. Conversely, when C/N is high, the growth of microbes is limited by N deficiency (Wu 2003). At present, the denitrification process mainly consists of heterotrophic denitrification, with C sources serving as electron donors. Lorrain et al. (2004) compared denitrification rates when using different sources of carbon. The result showed that sucrose-supported denitrification required more carbon sources in order to reach the peak, optimal processes. Weier et al. (1993) studied similar results by incubating intact soil cores in the laboratory and concluded the denitrification rate is limited by shortages of C. Simultaneously, Li et al. (2020) used the functional gene microarray to quantify the relative abundance of microbial nitrogen functional genes and showed

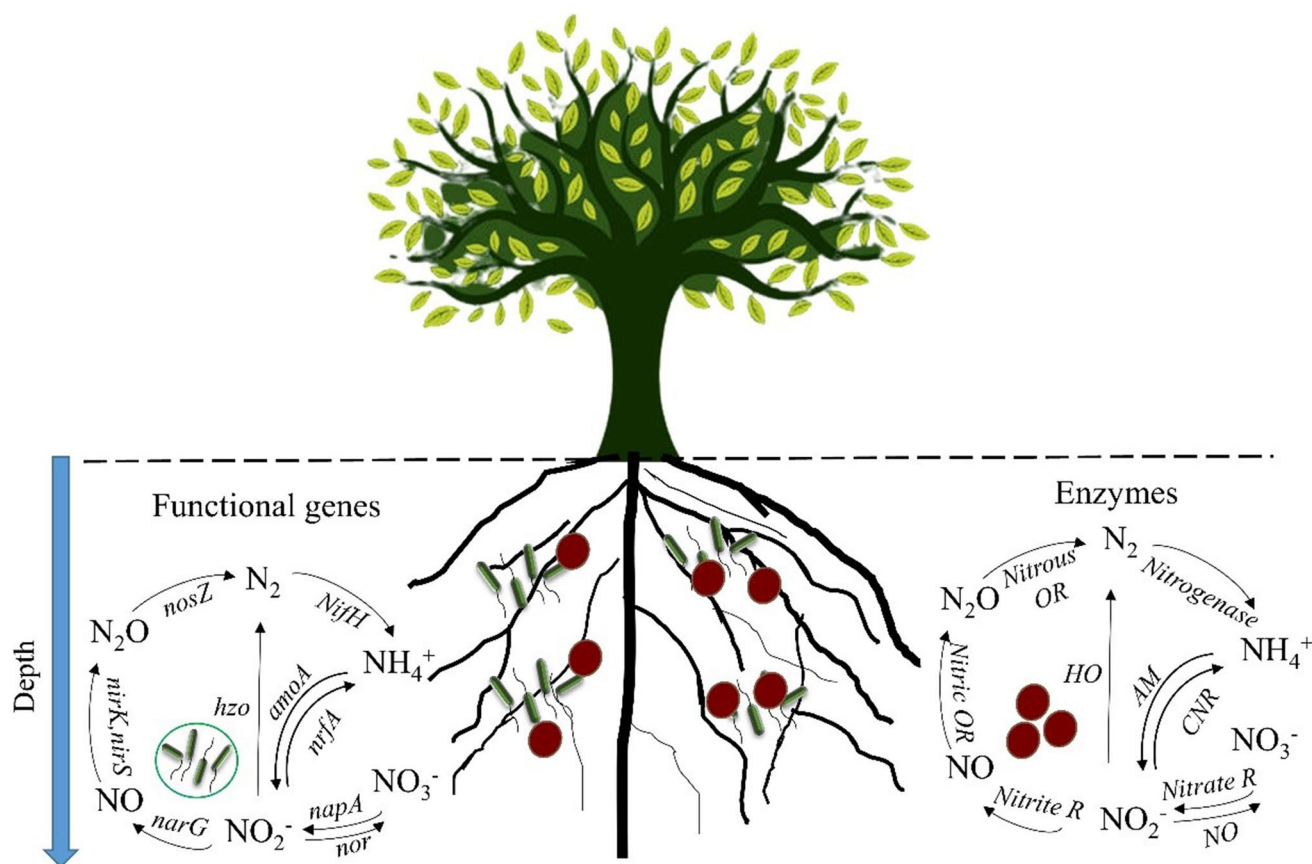


Fig. 3 The main microbial functional genes and enzymes involved in the N cycle in the plant-soil system

that the relative abundance of nitrogen functional genes was significantly reduced by high-concentration N treatments. Moreover, plants can indirectly affect N transformation through competition for N with microbes in the soil, and higher plant N uptake rates decrease soil N availability with consequences for the abundance and/or activity of N-cycling microbes (Moreau et al. 2019). In relation to oxygen, Xu et al. (2020) indicated that optimal soil O_2 conditions can improve soil N cycle and N transformation. Studies on soil N transformation by microorganisms have mainly focused on natural factors, but recent evidence showed that human activities such as the application of pesticides (Díaz-López et al. 2019), antibiotics (Zhi and Zhang 2019), and emerging pollutant accumulation in the soil have a significant impact (Ke et al. 2020). Therefore, we need to further understand the relationship between functional microbes and their influencing factors on N cycles as well as their transformation processes before we can optimally employ functional rhizosphere microbes in plant-soil systems.

Overall, the influence of plants on the migration and transformation of nitrogen in plant-soil systems is complicated and uncertainties remain about the mechanisms by which root exudation affects microbial N transformation

and supply dynamics of N to plants. Despite early studies highlighting the role of plant roots, plant metabolism, root exudates, and rhizosphere-microorganisms' interactions for the N migration and transformation, major uncertainties exist about this interdependent relationship.

4 Research Opportunities

4.1 The Quantification of N Migration and Transformation Pathways by Deep-Rooted Plants

Since the 1960s, understanding and quantifying the interaction between plants and soil to explore the role of plants in the N cycle has attracted widespread attention. Although the focus on studying the rhizosphere is increasing, the main focus is still on its relation to crop characteristics. Further research should investigate how the plant density and root depth of trees and shrubs affect N migration and transformation. Existing research mainly focuses on the influence of biomass indicators such as root activity, root

length, and root volume on soil preferential flow. There are few studies on the effect of various root structures on the soil preferential flow for single or multiple plant populations. Additionally, there are several major limitations in the quantification of root characteristic effects on N migration. As a result, available data is not sufficient to explain the relationship between plant root architecture and N migration in a plant-soil system. Further studies should aim to specifically investigate the quantification of N migration and transformation pathways by deep-rooted plants through isotope tracing and molecular biology techniques, combined with model simulation and other technical means.

4.2 The Effect of Environmental Stress on the N Migration and Transformation in the Plant-Soil System

The relationship between plants and nitrogen-related microbes varies with the seasons and plant growth. Factors such as climate change and rainfall characteristics cause highly variable water fluctuations in soil. For example, alternative dry and wet characteristics may cause irregular to extreme water stress on plants and rhizosphere microbes (Bagheri Novair et al. 2020). Investigation of the seasonal and interannual changes of plants in soil is essential to characterize the influence mechanism of plants on N migration and transformation. It can also help to predict how plants will function under the increasingly significant impact of climate change. It is worth noting that most researchers

conduct this research during summer and early autumn, while less conduct research during late autumn, winter, and spring. Future research should address this knowledge gap and strive to conduct seasonal experiments for a more complete and robust understanding of plants and their relationship with the mitigation and transformation of nitrogen. Furthermore, the majority of the research spans only 2 years. As a result, further studies on the impact of age on vegetation performance, especially for plants, as well as its relationship to nitrogen should be done over the course of > 2 years.

Apart from single-factor environmental stress, the major challenges of the quantification of N migration and transformation in a plant-soil system may also originate from the multiple interactions between biological and non-biological factors. Possible solutions include establishing N migration and transformation models for each factor. For example, an integrated migration and transformation model based on microbial action, plant uptake, sediment release, and substrate adsorption was established to evaluate the temporal-spatial distribution of N in the wetland (Li et al. 2021). However, it remains unclear how to up-scale single factors into multiple or interrelated factors under effective parameters. Additionally, the influences of different kinds of plant-soil systems on N migration and transformation processes vary widely and require many studies on different soil types and plant species to draw reliable conclusions on the combined influence of soil properties, plant types, climate conditions, and microbe activities.

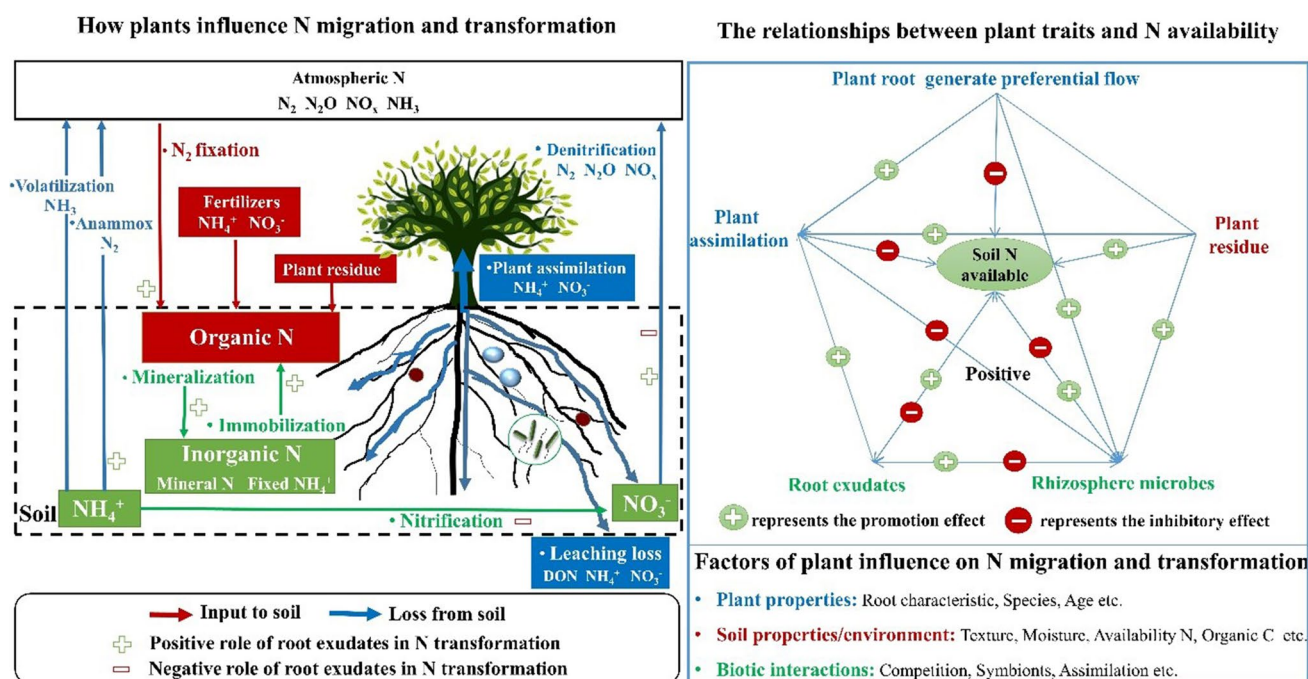


Fig. 4 A schematic of the influence of plants on the migration and transformation of nitrogen in plant-soil systems

5 Conclusions

The advances concerning the N cycle mechanism in the plant-soil system and the factors controlling N migration and transformation are reviewed in this article. It can be concluded that many mechanisms exist by which plants can influence the N migration and transformation and it is significant to understand how the relationships among the plant roots, plant metabolism, root exudates, and rhizosphere-microorganisms interactions exist (Fig. 4). As is evident from our review, for N migration, plant roots particularly coarse roots (> 5 mm) and decayed roots help generate preferential flow paths readily in soil and accelerate N migration, which increases the risk of N pollution in groundwater by reducing the retention of N around plant roots and fertilizer utilization. Plants affect the N transformation process in multiple ways while affecting the N migration process. We highlight the main contributions of plants in N transformation as follows: (a) Residual NH_4^+ and NO_3^- in the soil can be eliminated by plant assimilation but returned to the soil through the decomposition of plant residues; (b) N transformation processes controlled by microbes can be impacted by certain root exudates released from plant roots; (c) plant rhizosphere functional microbes possess N functional genes encode enzymes that are direct indicators of the biological ability of N transformation, and carbon-to-nitrogen ratios.

However, there are major knowledge gaps regarding quantitative pathways of N migration and transformation by deep-rooted plants as well as environmental stresses, particularly the comprehensive influence of multiple factors such as meteorological conditions (e.g., rainfall, humidity, and temperature), soil conditions, and human activity on plant-soil systems. These gaps warrant future research on the mechanisms of N migration and transformation in soils with the presence of plants. Moreover, it should be acknowledged that existing methods for detecting preferential soil flow, plant assimilation, root exudates, and rhizosphere microorganisms can be further improved by a combination of isotope tracing and molecular biology techniques, as well as other technical means for increased accuracy. We advocate establishing a database about different soil types and plant species to advance and complete the knowledge about the quantification influence of N migration and transformation in plant-soil systems for further applications.

Author Contribution Bei Zhang has been involved in drafting the manuscript or revising it critically for important intellectual content. Dr. Liang Chen has made substantial contributions to the conception and design. Prof. Song Jin and Prof. Qizhong Guo have modified the English grammar and given final approval of the version to be published. Jie Hou has contributed to the tables and artwork correction. All authors read and approved the final manuscript.

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Declarations

Conflict of Interest The authors declare no competing interests.

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